Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

Proposer	Name	Prof. Wenjie Zhou	Email	wjz@illinois.edu
Institution / Department	*Department of Materials Science and Engineering, Department of Mechanical Science and Engineering (affiliate)* *University of Illinois Urbana-Champaign*			
Address	Materials Research Laboratory, 104 S Goodwin Ave MC-230, Urbana, IL 61801, USA			

2. Technology Overview

Title of Proposed Technology	*AI-Driven Development of Advanced Metallic Textiles: Converting Standard Wire into Smart, Impact-Absorbing Safety Materials*			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display ( ), Secondary Battery (√), Advanced Mobility (√), Next-gen Nuclear Power ( ), Advanced Bio ( ), Aerospace/Marine (√), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing (√), Quantum ( ) *(Please choose at least one of the areas above)*			
Industrial Technology Classification [1]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Mechanical・Materials		*Mechanical・Materials*	
	Medium Category	Small Category	Medium Category	Small Category
	Automobile / Rail Vehicles	*Safety Improvement Technologies*	*Robotics/Automation Machinery*	*Robot Design Technologies*
Concept of Proposed Technology	*This Technology Proposal introduces our capability to create advanced metallic textiles that act as "smart safety materials." Instead of relying on new alloys, we improve performance through* ***architecture****, meaning how wires and contacts are arranged so the material can absorb impact, reduce vibration, and protect components while staying lightweight. A key feature is a simulation-first development workflow that combines physics-based modeling with AI acceleration. In simple terms, we can rapidly test many design options on a computer before building prototypes, which reduces cost and development time. Under the hood, the design framework is grounded in symmetry concepts from crystallography and in network models that predict how contact points carry force through the textile.* *Importantly, this program is designed to support two collaboration modes for Korean SMEs:* *(1) Application-first mode, where we improve performance using the SME's existing manufacturable mesh products and focus on integration and product validation.* *(2) Co-design mode, where we jointly develop new architectures if the SME is interested and able.*			

(max 10 pages)

Rationale for Joint R&D Project	*Existing Korean SMEs (e.g., Jinyoung Korea) already have specialized wire-mesh manufacturing capabilities, but face a practical challenge: changing a mesh design can require major equipment investment, and some companies prefer to focus on new applications for existing products rather than new structural designs. This joint project is built to be valuable under real industrial constraints. UIUC contributes a fast, simulation-first workflow and standardized testing to quantify impact absorption, vibration damping, durability, and integration performance. The Korean SME contributes manufacturing knowledge, product engineering, and access to real application environments (mobility, battery packs, robotics, aerospace/marine). Collaboration is essential because the fastest route to commercialization is not only better structure, but also the ability to integrate the material into real components, meet quality requirements, and validate performance in realistic use conditions.* *This proposal is intentionally written to be non-exclusive so multiple SMEs can see clear entry points, whether they manufacture wire mesh, metal textiles, forming/press components, battery modules, robotic systems, or transportation hardware.*
Keywords	1. *Metallic textiles* 2. *Impact absorption, shock protection, and vibration damping* 3. *Contact networks and topological mechanics* 4. *Scalable manufacturing*

(max 10 pages)

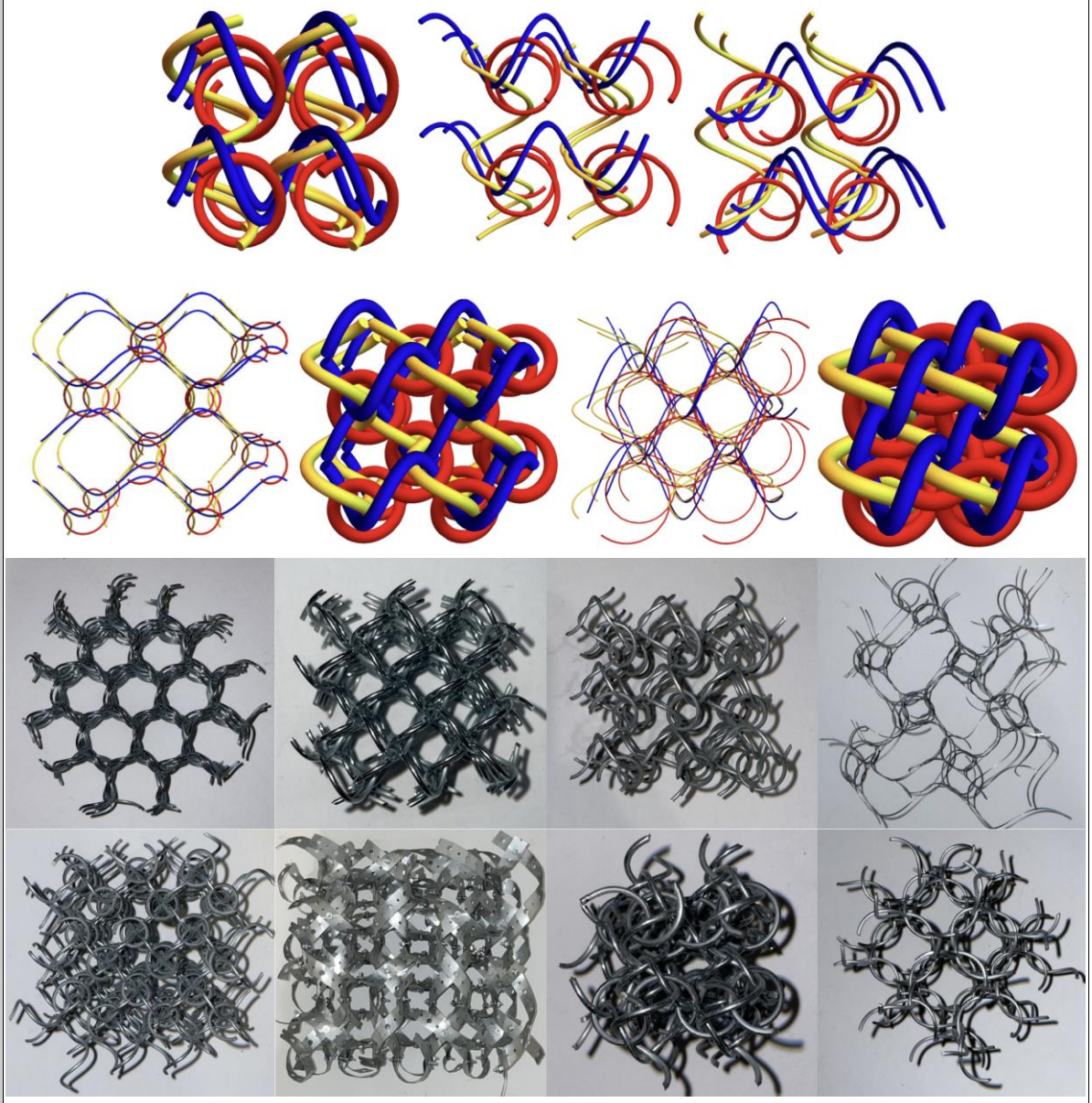
3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (3)	End of the Project	Level (6) [with a pathway to TRL 7 via customer reliability testing depending on partner access]
R&D Objectives	*This project will advance two complementary development tracks so that SMEs with different capabilities can participate.* ***Track I: Application-first development (for SMEs that prefer not to change the mesh design)*** • *Establish performance baselines for SME-provided metallic textile products and define target application requirements.* • *Develop integration methods such as stacking, pre-compression, boundary constraints, joining, packaging layouts, and surface protection approaches as needed for the application.* • *Produce prototype components and validate in relevant tests (impact, vibration, durability, environmental exposure as applicable).* • *Deliver a product-facing "engineering package" that includes performance maps, test protocols, and integration guidelines.* ***Track II: Co-design development (for SMEs interested in new architectures)*** • *Use simulation-first exploration to generate manufacturable architecture candidates within the SME's constraints.* • *Evaluate designs based on performance and manufacturability, then prototype and validate.* • *Create a manufacturable design window and quality metrics.* • *Focus on design changes compatible with existing tooling / low capital modification*			
R&D Target Specifications	***1) Demonstrator targets*** *A. Advanced Mobility (safety + NVH): shock/vibration control inserts, mounts/isolators, crash and drop protection.* *B. Secondary Battery (module safety infill): pack liners/spacers, crush/impact protection infill, vibration protection.* *C. Robotics & Manufacturing: protective skins/guards, end-effector protection, vibration protection for automation equipment.* *D. Aerospace/Marine: shock and vibration isolation for harsh environments and lightweight protection.* ***2) Performance targets*** *Impact/shock* • *Peak force reduction: 30–60% vs SME baseline at comparable thickness/mass* • *Specific energy absorption: 1.5–3× vs baseline* • *Rebound control: 30–50% reduction in rebound energy/height* • *Desired response: stable, progressive crushing (avoid brittle collapse)* *Vibration/NVH* • *Transmissibility reduction at resonance: 30–60% vs baseline* • *Damping improvement: ≥50% improvement in loss factor or equivalent metric* • *Validate improvement in at least one agreed frequency band (typical starting ranges: 20–500 Hz mobility, 10–300 Hz robotics, 10–2000 Hz aerospace)* *Durability* • *Retain ≥80% of key performance metric after 1,000 cycles (early) and target 10,000 cycles (late-stage, if applicable)* • *Quantify and minimize permanent set/thickness loss* *Environment (as relevant to application)* • *Mobility: temperature and humidity screening, basic fluid exposure if needed* • *Battery: vibration + crush; optional electrical isolation/barrier solution if required* • *Marine/aerospace: corrosion screening (salt/humidity) as needed*			

(max 10 pages)

	3) Manufacturing and integration targets *Integration (Track I): demonstrate major gains using SME's existing mesh by tuning confinement, pre-compression, stacking/graded layers, packaging, and attachment (minimal equipment change).* *Manufacturability (Track II): prototype within defined process constraints; target thickness tolerance ±10% early, ±5% late; areal density tolerance ±5%; define simple QC metrics (mass/area, thickness, stiffness window, uniformity).* ***4) Deliverables*** 1. *Performance Map (thickness/density/confinement/stacking → shock & vibration metrics)* 2. *Integration Playbook (packaging/attachment designs by application)* 3. *Test Protocol Kit (standardized methods + acceptance criteria)* 4. *Prototype Demonstrator Package (component-level build + validation data)*
R&D Plan	***Phase 1*** *WP0 (Month 0–1): Requirements & selection* • *Choose target applications, define success metrics, confirm manufacturing constraints* *Milestone M1: agreed requirement sheet + test matrix* *WP1 (Month 1–2): Baseline benchmarking* • *Test SME baseline mesh and incumbent alternatives; identify bottlenecks (rebound, densification, low damping, fatigue)* *Milestone M2: baseline performance report* *WP2 (Month 2–4): Retrofit integration prototypes (Track I)* • *Improve performance without changing mesh model via confinement frames, pre-compression, stacking/graded layers* *Milestone M3: at least one configuration shows ≥30% improvement in a key metric vs baseline* *WP3 (Month 2–5): Co-design screening (Track II)* • *Simulation-first down-select of manufacturable candidates within SME constraints* *Milestone M4: go/no-go decision for new-architecture prototyping* *WP4 (Month 5–6): Phase 1 demonstrator + Phase 2 plan* • *Build 1 component demonstrator per selected application class; define Phase 2 milestones and commercialization plan* *Milestone M5: demonstrator + complete Phase 2 workplan* ***Phase 2 (Tentative, Months 7–36; 30 months)*** *WP5 (Months 7–18): Optimization + QA readiness* • *DOE on key parameters (thickness/density/confinement/stacking); repeatability across builds; define QC acceptance windows* *Milestones:* • *M6 (Month 12): repeatable prototypes meeting tolerance and performance targets* • *M7 (Month 18): finalized component design ready for pilot runs* *WP6 (Months 12–28): Application integration + environment screening* • *Mobility: shock + NVH + thermal/humidity screening* • *Battery: crush + vibration; optional insulation/barrier integration* • *Robotics: impact/abrasion screening* • *Marine/aerospace: corrosion + broadband vibration screening* *Milestone M8 (Month 24): component meets application-relevant targets + durability screening* *WP7 (Months 24–36): Pilot-scale demonstration + customer-ready package* • *Pilot manufacturing runs, yield/tolerance reporting, reliability plan aligned with customer needs; prepare demo kits and spec sheets* *Milestone M9 (Month 36): pilot prototypes + clear product launch pathway*

(max 10 pages)



(max 10 pages)

4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• *Provide simulation-first modeling and AI-assisted screening to accelerate development while respecting partner manufacturing constraints.* • *Design and execute standardized mechanical testing and failure analysis (impact, vibration, cyclic durability).* • *Translate results into practical engineering guidance: performance maps, integration rules, recommended configurations.* • *Support prototype design iterations and validation planning for Phase 2.* • *Manage publication and IP strategy carefully so public disclosure does not reveal confidential design details.*
Expected R&R of Korean SME Partner	***For Track I (application-first)*** • *Provide manufacturable metallic textile products, components, or samples.* • *Provide application context, integration constraints, and target performance requirements.* • *Support prototype fabrication using existing processes and assist in component integration.* • *Support validation tests relevant to the target market (mobility, batteries, robotics, aerospace/marine).* • *Participate in commercialization planning, customer engagement, and scaling strategy.* ***For Track II (co-design)*** • *Share manufacturing constraints and feasible process windows in a non-confidential way.* • *Prototype selected new architectures and support scale-up feasibility evaluation.* ***Illustrative example (non-exclusive):*** *A wire-mesh or metallic textile company (e.g., Jinyoung Korea) that has established production equipment but prefers to focus on new applications rather than investing in new design tooling is a good fit for Track I.*

(max 10 pages)

5. Expected Outcomes/Impact

Expected Outcomes and Impact	***Technical outcomes*** • *Validated, application-ready metallic textile safety solutions with quantified impact absorption, vibration reduction, and durability.* • *A public-safe "performance and integration library" that helps Korean SMEs adopt advanced metallic textiles without requiring deep theoretical expertise.* • *Prototype demonstrators in one or more strategic areas: advanced mobility safety and NVH, battery module protection and safety infill, robotic protection and manufacturing safety, and aerospace/marine vibration and shock control.* ***Commercialization potential*** • *Near-term adoption through retrofit integration into existing products and platforms.* • *Clear pathways to product differentiation based on measurable performance rather than new materials.* • *Faster development cycles using simulation-first iteration, reducing prototyping cost and time.* ***Korean benefits*** • *Strengthens Korean SMEs' competitiveness in global markets for safety materials, mobility components, battery safety components, and robotics manufacturing.* • *Builds domestic capability in design-to-manufacturing workflows and standardized validation methods.* • *Supports new exportable products that align with Korea's strategic technology priorities.*

6. References

(1) **Zhou, W.***, Nadarajah, S., Li, L., Izard, A.G., Prachet, A.K., Patel, P., Yan, H., Xia, X., and Daraio, C.*, 3D Polycatenated Architected Materials, ***Science***, 2025, *387*, 269 [Cover Article].

(2) Yan, H., Meng, Z., **Zhou, W.** *, and Daraio, C.*., Rigidity Criteria for Chainmail Consisting of Tessellations of Torus Knots, ***Physical Review Letters***, 2025, *135*, 088201.

(3) Wetzel, D., Gailiunas, P., Gaither-Ganim, M., and Holt, W.,Triply Periodic Helical Weaves. *arXiv:2402.07849.*

(max 10 pages)

☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (Appendix 1, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
	Basic Research Stage (1-2)	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

--- 10-page limit ---